🛠️ MEC 427 Advanced Sensors and Actuators Challenges 1-5

📅 Date & Time

- **Report Date:** `11-22-2025`
- **Time Block:** `11-17 - 11-22`
- **Time Spent This Session:** `8 hours / 30 minutes`
- **Cumulative Time on Project:** `33 hours / 30 minutes`

🗂️ Project Overview

- **Project Title:** `Challenge 4: Cable trolly, and Challenge 3: Water control system.`
- **Project Objective(s):**
- `Design an actuator system that can traverse a steel tension cable.`
- **Current Phase:** `Design Phase`

🔧 Task / Activity Details

- **Module / Subsystem:** `Challenge 4 Gantry motion control system design`
- **Specific Task or Activity:**
- **Time Spent on This Task:** `6 hours / minutes`
- **% of Total Time:** `80%`
- **Resources Used:**
  - **Hardware:**
  - **Software:**
  - **Tools:** `Grok AI, Fusion 360.`
  - **Documentation:** `Design Sketches CAD designs`

- **Module / Subsystem:** `Challenge 3 Water control system coding and testing`

- **Specific Task or Activity:**

- **Time Spent on This Task:** `2 hours / 30 minutes`

- **% of Total Time:** `50%`

- **Resources Used:**

  - **Hardware:** `Adafruit CPX, capacitive touch sensors, servo pinch valve, water tower model.`
  - **Software:** `VsCode, CircuitPython.`
  - **Tools:** `Copilot AI`
  - **Documentation:** `AI conversation.`

- Project Status Update:
- **Task/Module:** `Challenge 4 Gantry motion control system design`
- **Project Completion Status:** `75% complete`
- **Milestones Reached:** `Finalized CAD design for gantry motion control system.`
- **Blockers or Issues:**
-
- **Task/Module:** `Challenge 3 Water control system coding and testing`
- **Project Completion Status:** `8% complete`
- **Milestones Reached:** `System is able to read water levels and actuate the valve based on sensor input.`
- **Blockers or Issues:** `Fine-tuning the sensor calibration to avoid false readings.`

📦 Results & Deliverables

- **What Was Achieved:** `Completed Challenge 4 Design and finished the first phase of coding for challenge 3.`
- **Artifacts Created:** `CAD designs for cable trolly system. Code and test results for water control system.`
- **Validation / Testing Outcomes:** `Water tower actuation tests show reliable operation with current calibration.`
- **Changes Made to Design or Approach:** `GroK AI was used to assist in design ideation and STL generation with tinkercad.`

📜 Narrative & Observations

> *Use this section for detailed descriptions, reflections, and context. Include technical notes, emotional observations, and rationale for decisions.*

- **Narrative Summary:** `Challenge 4 required several more iterations of design before I settled on the 'sandwich' concept. Using Grok AI to brainstorm design ideas and Fusion 360 for CAD modeling greatly expedited the process. The idea didn't really come together until after researching cable and zipline systems in industry applications. For Challenge 3,I hit the limitations of the CPX memory when implementing more features. Copilot was able to streamline the code and reduce memory usage significantly. Encountered power spikes when the servo actuated, which I mitigated by adding a separate power supply for the servo. Learning a lot about electronics, wiring, and power management. My lack of experience in engineering is a recurring challenge, but I'm making progress.`
